**Raeyaan Muppaneni (https://raeyaanmuppaneni.github.io/)**

## EDUCATION

**Irvington High, Fremont, California** (August 2024 - Present)
**Happy Academy, San Jose, California** (August 2023 - Present)
**Basis Independent Silicon Valley, San Jose, California** (August 2023 - June 2024)
SAT: 1540 800M/740RW; PSAT: 1500/1520
GPA: 4.00/4.0 (unweighted) 4.74/5.0 (weighted)
National Merit Semifinalist; AP Scholar with Distinction

## AP (Scores) and HONORS Courses

AP US Government and Politics (5), AP Computer Science Principles (5), AP Physics I (5), AP Calculus AB (5), AP Macroeconomics (5), AP World History (4), AP English Language and Composition (4), AP Calculus BC (5), AP Computer Science A (5), AP Physics C: Mechanics (5), AP Physics C: Electricity & Magnetism (4), AP Psychology, AP Environmental Science, AP Statistics, AP Chemistry, AP Seminar, AP Research, AP Spanish, Chemistry Honors, English 10 Honors, Precalculus AB

## Additional Coursework

Calculus 2, AI Machine Learning, Computer Architecture, Data Visualization, Multivariable Calculus, Quantum Physics, Data Science, Linear Algebra, AC DC Circuits

## RESEARCH PUBLICATIONS AND PATENT

**Publication:**

- **Real-Time Control of a Low-Cost Robotic Arm Using EMG Signal Classification by AI-Based Machine Learning on Raspberry Pi** (https://ieeexplore.ieee.org/document/11496321). Published in IEEE Xplore, proceedings of IEEE WcCST 2026 (co-sponsored by the IEEE Computational Intelligence Society), March 2026.
- **VITALIS: A Patient-Owned Agentic Framework for Emergency Detection and Governed Clinical Disclosure** (https://www.youthinnovationjournal.org/researchlib/papers/vitalis). Co-author. Published in Youth Innovation Journal, Summer 2026.

**Patent:**

- **U.S. Provisional Patent Application No. 63/887,594.** AI emergency-detection wearable that monitors vitals in older adults and chronically ill patients and escalates to caregivers, clinicians, and emergency services; co-designed with Robolabs.

## RESEARCH AND ENGINEERING PROJECTS

**Pressure-Sensing Insole for Para-Athletes (Stanford SIMR and Human Performance Lab)**
**(**https://github.com/raeyaanmuppaneni/Blade-Insole**)**- *Electrical Engineer, Team Lead*
June 2026- Present (Grade 12) ; 12hrs/wk, 25wks/yr

- Selected for the Stanford Institutes of Medicine Summer Research Program (SIMR) with less than 5% acceptance rate; project continues in the Stanford Human Performance Lab
- Designing a capacitive pressure-sensing insole that tracks gait in blade-running para-athletes

- Found the need by interviewing Sydney Barta, a Stanford para-athlete: blade runners cannot use standard lab tools such as instrumented treadmills
- Built the full rigid-board schematic (FDC2214 capacitance-to-digital converter, LC tank circuits, OPA4354 active shield drivers, STM32 Nucleo) through multiple TA review rounds
- Hand-soldered SMD parts on boards and got data from all 8 sensors; presented at the SIMR poster session
- After SIMR, leading the team's continuation under PhD student and mentor Michelle Joyce: setting next steps, running progress updates, scheduling lab time
- Redesigned the rigid PCB to mount directly on the microcontroller, coordinated the flex-board redesign, ordered version 2 boards, and got stalled firmware running within days
- Currently building the enclosure and calibrating device against force plates; entered the team into the Conrad Challenge

**EMG Robotic Arm (**https://github.com/raeyaanmuppaneni/EMG-controlled-robotic-arm**)**- *Independent Researcher*
June 2024- March 2026 (Grades 10, 11) 6hrs/wk, 35wks/yr

- Built an EMG-controlled robotic arm for people with motor dysfunction and amputees
- Did research work under mentorship of former Stanford PhD candidate Jon Stingel
- Used raspberrypi4, EMG sensors, lithium ion battery, electrodes, robotic arm with servos and 4 degrees of freedom, code built on google colab
- Collected data from my own arm using the EMG device I built, which had a simple PCB and 3d-printed enclosure
- Tested different machine learning algorithms: k nearest neighbors, Multilayer Perceptron (MLP), Random Forest, and Support Vector Machine (SVM); testing different hyperparameters
- Was able to get 93% accuracy on test data with Random Forest model; got robotic arm moving with corresponding movements in actual arm
- Competed in Alameda County Science and Engineering Fair (ACSEF) got 2nd place award
- Wrote the research paper, published in IEEE WcCST 2026
- Did further research to use Jacobians and inverse kinematics to see how to optimize a robotic arm that minimizes jerk and power consumption
- Tested different trajectory planning methods such as cubic spline and linear interpolation

**AI Emergency-Detection Wearable (with Robolabs)**- *Codesigner and Coauthor*
August 2024-July 2026 (Grades 10, 11); 3hrs/wk, 45wks/yr

- Co-designed a wearable-based system to monitor vitals in older adults and chronically ill patients living alone and escalate to caregivers, family, clinicians, and emergency services
- Core of the design is an AI agent on the patient's own devices that learns the user's normal vitals, checks unclear readings before raising an alarm, and sends responders a short summary of what it observed
- Studied HIPAA emergency-access rules; designed each disclosure to be limited by recipient role, time-limited, and logged to balance patient privacy with safety
- Published paper on research in Youth Innovation Journal
- Filed provisional patent application number 63/887,594

**Collision-Avoidance Wearable for Blind Students**- *Designer and Builder*
August 2023 - April 2024 (Grade 9); 6hrs/wk, 35wks/yr

- Built object detection and classification device for blind people to prevent collisions
- Used YOLOv8 pretrained on an 80-class dataset and additionally implemented my own trained crosswalk dataset

- Used raspberrypi4, camera, speakers and designed casing for device
- Competed in Synopsys science fair; got 2nd place category award and Mayor's Award

**Assistive Devices for Neurodiverse Kids**- *Designer and Builder*
July 2026-Present (Grade 12); 3hrs/wk, 25wks/yr

- Building a handheld speech-button device for non-verbal children (ESP32-S3, 9 record-and-play messages, rechargeable battery); designed and ordered PCB
    - Started after a non-verbal student at a Rooting Minds workshop could not signal whether he understood the game
- Built an EEG-based app that tracks focus states for neurodivergent students and competed with it in Congressional App Challenge
- Currently working on making an app where the physical card game can be played to increase accessibility to neurodivergent children globally

**Evodyne Robotics**- *Engineer*
July 2025 - May 2026 (Grade 11); 5hr/wk, 35wks/yr

- Built advanced robotic dog with 3d printed parts, arduino, and raspberry pi
- Worked on actuator for humanoid, working on Fusion 360 for design

## VOLUNTEERING, LEADERSHIP, AND ACTIVITIES

**Rooting Minds (501c3 nonprofit)**- *President and Cofounder (11th & 12 Gr)*
June 2025 - Present (Grades 11, 12); 8hr/wk, 30wks/yr

- Co-founded with brother to teach problem solving, science, and creative thinking to neurodivergent children
- Designed learning games, including an animal and math matching card game with clues or problems to make learning fun and improve cognitive ability
- Designed, produced, and distributed 100 learning game card decks
- Leading biweekly workshops in partnership with Friends of Children with Special Needs (FCSN), impacting 150+ students
- Worked with students in WeEMBRACE, a nonprofit for neurodivergent children, running online classes to teach art and make kids play our card game
- Paired 10+ volunteers with kids for one-to-one workshops and socials
- Building an app with digital versions of the games to increase accessibility; building a speech-button device for non-verbal students
- Bringing more volunteers to Irvington High's Neurodiversity Club

**Arnold & Marsden Math Olympiad Circle (AMMOC)**- *Math Competitor*
June 2023 - Present (Grades 9, 10, 11, 12); 15hrs/wk, 45wks/yr

- Studied number theory, geometry, combinatorics, and algebra in Arnold & Marsden Math Olympiad Circle (AMMOC)
- Competed in 30+ competitions across group and individual
- Qualified for USAMO (only a couple hundred students a year); 144 AMC 12B, 13/15 AIME
- Worked through various books like Problems and Solutions in Mathematical Olympiad High School 1 & 3, Challenging Problems in Geometry, A Gentle Introduction to the American Invitational Mathematics Examination

**Robotics Club: Basis Independent Silicon Valley**- *Mechanical Engineer and Driver*
August 2023 - April 2024 (Grade 9); 7hrs/wk, 35wks/yr

- Built drivetrain and designed catapuncher for robot
- Competed in several robotics competitions
- Made it to semifinals out of 200+ teams, qualifying for regionals
- Made it to eliminations out of 250 teams in regionals
- Practiced driving the robot and was driver during competition

**Cricket (American School of Cricketing Excellence)**- *Team Captain*
July 2019- June 2025 (Grades 9, 10); 12hr/wk, 40wks/yr

- Captain of the Under 15 team: came up with strategies and supported teammates
- Won 7 tournaments in LA and Bay Area
- Traveled across the country for tournaments such as Oregon
- Qualified for USA Cricket intrazonals, barely missing out on zonals, the highest level
- 10+ player of the match awards

## Internships

**Fluora Bioscience Laboratories**- *Electronics Engineer Trainee*
Fall 2026 (Grade 12); 6hrs/wk, 6wks/yr

- Designing the electronics for a portable fluorometer (fluorescence spectrometer) at a biotech company
- Weekly design reviews with manager and separate weekly meetings with whole team
- Leading the electronics breadboarding team
- Button-triggered measurement using alternating blue and green LEDs, LED irradiance sensing, and two switchable optical filters
- Photodetector and amplifier chain sampled at 100 Hz; measuring difference in light intensity of emitted and absorbed light
- Worked on Arduino code with esp32 microcontroller
- Rechargeable battery (USB or wireless) sized for 100+ measurements per charge

## AWARDS/RECOGNITION

- USA Mathematical Olympiad (USAMO) Qualifier (2026)
- 144 on AMC 12B
- 13/15 on AIME
- National Merit Semifinalist
- AP Scholar with Distinction
- Congressional Bronze Medal and Gold Certificate
- High Distinction Australian Mathematics Competition 2025
- 2nd place Synopsys Science and Technology Championship (2024)
- 2nd place Alameda County Science and Engineering Fair (2025)
- Recognition Award from the Mayor of Fremont
- Bronze American Mathematics Olympiad 2024
- Bronze ASIAROPE 2023
- Distinction in Cayley Contest 2025
- Distinction in Fermat Contest 2026
- Performance with Honors in COMC 2024
- Finalist in International Youth Math Challenge IYMC
- Math League International Summer Challenge 2024: Bronze
- "Solved Problem in Math Reflection Journal" recognition

- Caribou Contest: Top 15% (November, January), Top 25% (October)
- Robotics: Semifinalist at a rookie competition; elimination rounds at a regionals competition
- Cricket: 10+ Player of the Match awards; 7 tournament wins
- Certificate of Honors in BASIS Independent Silicon Valley
- Distinguished Mathematics Scholar presented by Happy Academy
- Technology for Good Award presented by Happy Academy

**SKILLS/TALENTS**

- PCB Design: KiCad, EasyEDA Pro, soldering, reading chip datasheets
- Embedded Hardware: STM32, ESP32-S3, raspberry pi, sensors, circuits
- Programming and Design: Python, Java, C++, Fusion 360
- Languages: English, Telugu, Spanish (taking AP Spanish)
- Athletics: Calisthenics, cricket, basketball